The 16th e-ICON World Contest

SnapCare

H1 | SnapCare

01 The problem

A prescription is written to be read. In Korea **59.7%** of adults aged 65 and over fall below a sufficient level of health literacy.

Removing the text is not enough. Only 17 of 108 pictograms met the ANSI threshold in the Philippines. A set of *n* drawings says *n* things.

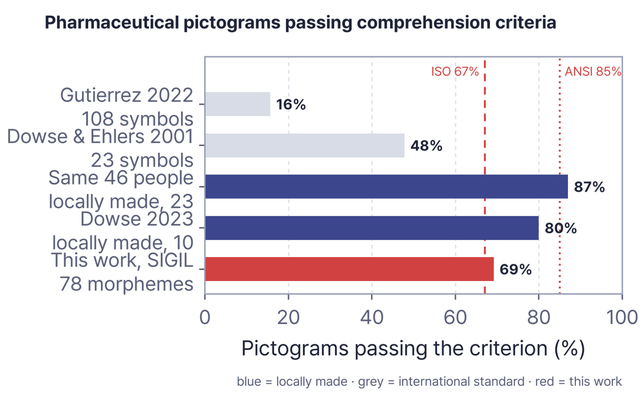


Figure 1. Locally made sets pass the criteria; international sets do not.

02 What must be true

At no point is the user required to read or type.

- Compose from a **grammar**, not a list
- Rebuild the sentence, not read the string
- Judge on the device — no account, no network

03 How it works

1. **Photograph.** Point the camera. Nothing is typed.
2. **Read on the device.** Everything is parsed in the browser; the photograph never leaves it.
3. **Compose.** SIGIL builds each frame from morphemes.
4. **Speak.** VOCA rebuilds the sentence the patient needs.

04 SIGIL — a grammar

A head stands on its own; a modifier goes into a slot inside a head. Five operators apply on top. The 78 morphemes were derived backwards from the distribution of medication error.

Element	n	Example
Head	57	pill_bottle
Modifier	21	time_morning
Operator	5	no{}, half{}
Morphemes	**78**	**composed, not selected**

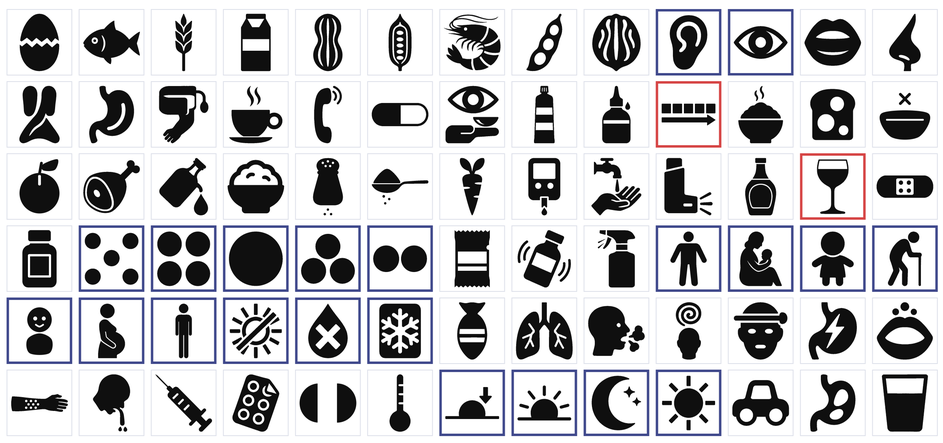


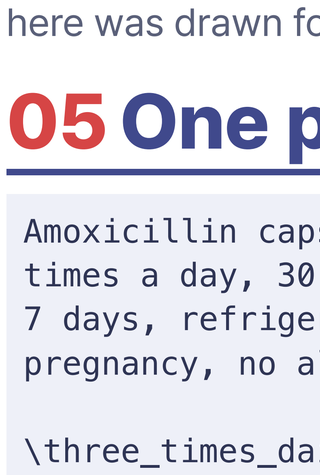
Figure 2. Every morpheme the engine holds. Nothing here was drawn for a particular prescription.

05 One prescription

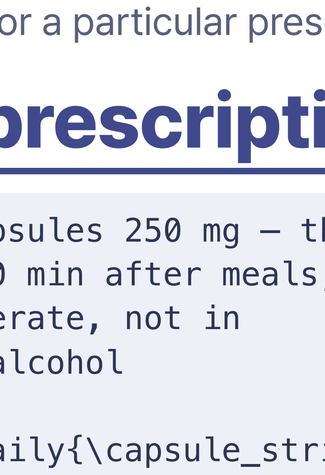
```
Amoxicillin capsules 250 mg — three
times a day, 30 min after meals,
7 days, refrigerate, not in
pregnancy, no alcohol

\three_times_daily{\capsule_strip{\food_bowl}}
\capsule_strip{\time_morning}
\capsule_strip{\storage_fridge}
\finish_course  \no_alcohol  \state_pregnant
```

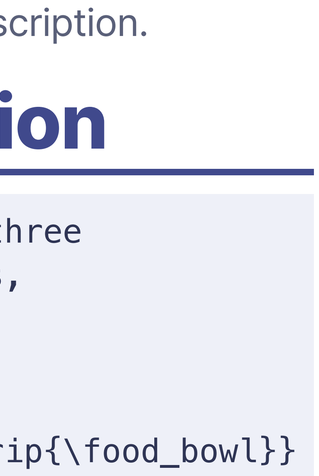
Ten frames from one prescription. **Four carry what was buried in the warning box**: finish the course, no alcohol, not in pregnancy, caution while breastfeeding.



a. Prescription



b. Food label



c. Setup

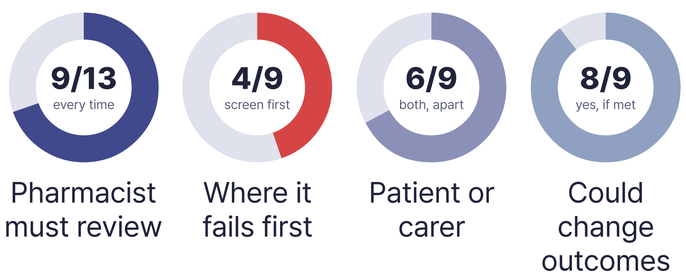
Figure 3. Every element on all three screens can be pressed to be read aloud.

06 What 39 experts said

1,103	39	22
experts written to, one letter each	answered (3.5%)	countries on five continents

> *The pictogram showed a pregnant woman where the prescription said not to, making the patient assume it is safe.*
>
> Prof. Atif Usman, Islamia University, Pakistan

Three respondents on three continents reported this, and it was fixed.



07 What we measured

Measure	Result
Formative runs 4 prescriptions × 7 conditions	**27 / 28** 0 / 28 before the fix
Semantic check of all 78 morphemes	**54** read as intended the 7 on screen were reworked
Nutrition model, Nutrition5k	**26.75%** mean MAE n = 507
Offline, server shut down	**Normal**

What we do not claim. No comprehension test with people has been run; ISO 9186-1 and ANSI Z535.3 must still be confirmed with low-literacy adults.

Try it now

`snapcare.fly.dev`

Runs in the browser. No install, no account, and it keeps working offline.

Host | Ministry of Education

Organizer | KEFA Korea Digital Education Frontiers Association